

Fact Sheet

COMMERCIAL AND CIVIL ROBOTIC RENDEZVOUS AND PROXIMITY OPERATIONS

June 2025

CONTACT: VICTORIA SAMSON
VSAMSON@SWFOUND.ORG



SECURE
WORLD
FOUNDATION

SUMMARY

Over the past two decades, several countries have begun developing rendezvous and proximity (RPO) technologies for non-military purposes such as satellite inspection, refueling, repairs, on-orbit assembly, and debris removal. In addition to government-funded programs, companies have also begun developing, testing, and operationalizing capabilities to provide commercial on-orbit satellite servicing capabilities such as life extension, repair, refueling, and inspection. In January 2020, the first commercial satellite servicing mission successfully docked with and repositioned a commercial communications satellite in the geosynchronous region. In early 2022, China successfully removed a dead satellite from GEO to the graveyard region. Several additional commercial/civil demonstrations and operational missions are planned over the next few years.

RENDEZVOUS AND PROXIMITY OPERATIONS (RPO) AND ON-ORBIT SERVICING (OOS)

Proximity operations are a series of orbital maneuvers executed to place and maintain a spacecraft in the vicinity of another space object on a relative planned path for a specific time duration to accomplish mission objectives. Rendezvous is a process whereby two space objects, artificial or natural bodies, are intentionally brought close together through a series of orbital maneuvers at a planned time and place. Taken together, RPO technologies enable a wide range of capabilities to support civil and commercial on-orbit satellite servicing (OOS) capabilities such as inspection, repair, refueling, assembly, and life extension. RPO capabilities can also be used for military and intelligence space activities such as intelligence, surveillance, and offensive weapons such as co-orbital anti-satellites.¹

CIVIL ROBOTIC RPO AND OOS TECHNOLOGY DEVELOPMENT AND TESTING

While RPO capabilities have a long history in human spaceflight, the past twenty years have shown a growing focus on developing robotic versions of these same capabilities. This focus includes government funding to conduct research and development of underlying technologies, technology transfer to industry, and industry efforts to commercialize the technology and identify customers.²

The first robotic RPO mission to be categorized as a civil operation was the November 1997 Japan ORIHIME and HIKOBOSHI test conducted by the Japan Aerospace Exploration Agency (JAXA). As part of the Engineering Test Satellite VII, the servicer satellite (Orihime) orbited the client satellite (Hikoboshi) and became the first successful instance of autonomous robotic rendezvous and docking with another space object.³ While orbiting, Orihime took pictures, inspecting Hikoboshi before redocking and finishing the autonomous test.

On April 14, 2005, NASA launched the DART satellite to conduct an autonomous rendezvous experiment

with a U.S. Navy communications satellite, MUBLCOM.⁴ DART ended up “bumping” into MUBLCOM during the test but both satellites were unharmed. An after-action report concluded that the incident was due to a malfunction of the automated rendezvous system, although not all details were released due to export control and classification restrictions.

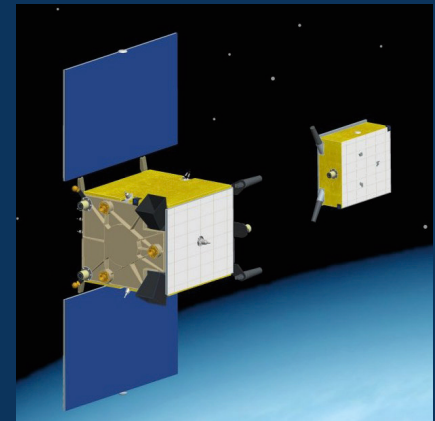
In 2010, Sweden began an RPO mission of their own focusing on autonomous guidance and navigation experiments.⁵ The pair of satellites in the PRISMA mission, PRISMA Target and Main, were also commonly known as Mango and Tango. This close-approach experiment also provided GPS navigation testing for future formation flying as well as other technology experiments for various European partners.

January 2016 marked the first university-led test of RPO capabilities. BEVO-2 and AggieSat4 were part of the second LONESTAR mission aimed at testing in-space navigation capabilities built and conducted by the University of Texas and Texas A&M.⁶ BEVO-2 was deployed from AggieSat4 after both were released from the ISS. The pair of satellites proceeded to practice communication and data exchange while station-keeping and updating their navigation. This marked a key capability to remotely conduct accurate communication transfer and flying controls with navigation and guidance techniques.

As part of their program to develop OOS capabilities to support future missions, NASA had two missions focused on on-orbit servicing, assembly, and manufacturing (OSAM), dubbed OSAM-1 and OSAM-2. OSAM-1’s primary mission was to demonstrate the ability to refuel a satellite, but also included a robotic arm known as SPIDER that would demonstrate the ability to assemble an antenna in space. However, NASA announced in March 2024 that it was canceling OSAM-1 due to it being years delayed and heavily over its budget, and in September 2024, NASA announced that it was wrapping up spending on the program by the end of the month.⁷ OSAM-2 involved a satellite called Archinaut One that would build, assemble, and deploy its own operational solar array; it was concluded in 2023 prior to a flight demonstration.⁸

Other commercial and civil RPO developments include Italy’s D-Orbit ION launched in September 2020, which offered rideshare opportunities to bring satellites to their operational orbits faster than typical drifting procedures.⁹ Additionally, Japanese start-up Astroscale’s ELSA-d servicer launched in March 2021 and successfully demonstrated the ability to rendezvous, inspect, and capture another object to support future active debris removal missions.¹⁰ The Swiss company ClearSpace originally had a contract from the European Space Agency to remove a 100-kg upper stage adapter from orbit,¹¹ but in 2024, the ClearSpace-1 mission was assigned a new target due to the original one hitting some debris and becoming too complex of a challenge. Planned for 2028,¹² ClearSpace-1 will rendezvous with and capture the ESA’s Project for On-Board Autonomy-1 (Proba-1) satellite, followed by a reentry in the Earth’s atmosphere.¹³

On October 24, 2021, China launched the SJ-21 satellite for a space debris mitigation mission.¹⁴ On December 25, 2021, the SJ-21 rendezvoused with the Compass G2, a defunct Chinese navigation satellite.



Prisma's Mango & Tango.

Image credit: European Space Agency



View of IS-901 from MEV-1 while docking.

Image credit: Northrop Grumman.

The Compass G2 had experienced a fragmentation event in 2016 which released six trackable pieces.¹⁵ The SJ-21 docked at some point and on January 21, 2022, used its onboard propulsion to pull the Compass G2 to an altitude above the geostationary belt.¹⁶

On February 18, 2024, Astroscale Japan launched the Adras-J satellite for a debris inspection demonstration mission. Between April 17 and November 30, 2024, Adras-J conducted close approaches, fixed-point observations, and fly-around observations with an H-2A rocket upper stage. On November 30, 2024, Adras-J successfully conducted a final close approach at a distance of 15 meters from the upper stage. Adras-J did not reach its

intended final distance because an autonomous abort was triggered by the satellite's onboard collision avoidance system.¹⁷

There have also been two commercial satellite servicing missions. The Mission Extension Vehicle 1 (MEV-1), a satellite developed by SpaceLogistics (a subsidiary of Northrop Grumman), docked with the Intelsat 901 (IS-901) satellite in March 2020 just above the geostationary belt and then brought it down to GEO to provide life-extension and repositioning services until April 2025, when it brought IS-901 back up to the GEO graveyard orbit and MEV-1 undocked.¹⁸ Its follow-up, MEV-2, launched in August 2020 and docked with Intelsat 10-02 in April 2021 in the geostationary belt to begin a similar five-year life extension mission.¹⁹

While there have only been a few commercial missions to date, the advances in technology, changes in national policy, and new regulations could spark a significant increase in the future. Recently, multiple start-ups have offered their first clients satellite servicing, active debris removal, and end-of-life services; this could be a promising future for the commercial space industry. The Consortium for Execution of Rendezvous and Servicing Operations (CONFERS), created in 2017 with initial funding from the Defense Advanced Research Projects Agency (DARPA), today has more than 80 members from several countries planning a wide range of future commercial OOS missions.²⁰

DUAL USE AND TRANSPARENCY

The growing advances in commercial and civil RPO and OOS capabilities raise important questions about the distinction between their activities and RPO conducted for military or intelligence reasons. Increased transparency is needed, along with the establishment of norms of behavior and technical standards, to increase the safety of commercial and civil RPO and OOS and help distinguish them from military and intelligence activities.

Summary of Known Commercial and Civil RPO & OOS Activities in Space

Country	Date of Activity	Servicer Satellite	Client Satellite	Launch Site	Notes
Japan	Nov. 1997	ORIHIME	HIKOBOSHI	Tanegashima Space Center	Demonstrated autonomous docking and satellite servicing with onboard mounted robotic system
USA	April 14, 2005	MUBLCOM	DART	Vandenberg AFB	NASA DART bumped into military satellite during proximity test, unclear if offensive test
Sweden	Aug. 2010 - July 2011	PRISMA Target	PRISMA Main	Dombarovskiy Air Base	Demonstrated autonomous guidance and navigation experiments
USA	Jan. 2016	BEVO-2	AggieSat4	Cape Canaveral	Pair of satellites deployed from ISS to practice station keeping and navigation
USA	Jan. 25, 2020 - Apr. 9, 2025	Intelsat 901	MEV-1	Baikonur	MEV-1 successfully docked to provide mission extension services for Intelsat 901 for five years
USA	June 15-29, 2020	Aerocube 10-B	Aerocube 10-A	Wallops	AC-10B entered a proximity orbit around AC-10A and used its on-board camera to take resolved images of AC-10A
Italy	Sept. 3 - Oct. 28, 2020	D-Orbit ION	Multiple	Guiana Space Center	ION performed rendezvous operations with multiple satellites to bring it to its designated orbit faster than normal drifting
USA	April 12, 2021-present	MEV-2	Intelsat 10-02	Guiana Space Center	MEV-2 successfully docked with Intelsat 10-02 to provide mission extension services
China	April 27, 2021 (launch)	NEO-1	?	Taiyuan Satellite Launch Center	NEO-1 planned to release a small piece of pseudo-debris and then capture it again using a net system. Mission status is unknown
Japan	March 2021 - May 2022	ELSA-D Servicer	ELSA-D Client	Baikonur	ELSA-D magnetically docked with the client sub satellite and demonstrated RPO stationkeeping and on-orbit inspection
China	Oct 2021 - Jan. 2022	SJ-21	Compass G2	Xichang Satellite Launch Center	SJ-21 docked with Compass G2 and towed it to the GEO graveyard, before detaching and returning to active GEO region
USA	Apr-23	LINUSS1	LINUSS2	Kennedy Space Center	One cubesat performed the role of a servicing vehicle for the other. The mission was used by Lockheed Martin to test various algorithms, SSA cameras, and the new ASPIN
Japan	Feb. - Nov. 2024	ADRAS-J	H-2A rocket upper stage	Rocket Lab Launch Complex	ADRAS-J successfully approached a rocket upper stage in LEO within 15 meters, which is the closest approach of space debris by a commercial space company to date. ADRAS-J reached a 15 meter distance on November 30, 2024.

ENDNOTES

1. NASA Goddard Space Flight Center, "On Orbit Satellite Servicing Study Project Report," October 2020, https://nexus.gsfc.nasa.gov/images/NASA_Satellite_Servicing_Project_Report_2010A.pdf.
2. Brian Weeden, "The Evolution of Space Rendezvous and Proximity Operations and Implications for National Security," September 6, 2018, <https://www.unidir.org/files/medias/pdfs/brian-weeden-presentation-eng-0-804.pdf>.
3. Mitsushige Oda, "On-orbit Satellite Servicing," JAXA, May, 2012, https://nexus.gsfc.nasa.gov/workshop_2012/oda_final_presentation_2012_workshop.pdf; "About Engineering Test Satellite VII 'KIKU-7,'" JAXA, accessed July 1, 2020, <https://global.jaxa.jp/projects/sat/ets7/index.html>.
4. "Overview of the DART Mishap Investigation Results," NASA, accessed July 1, 2020, http://www.nasa.gov/pdf/148072main_DART_mishap_overview.pdf.
5. Stephen Clark, "Sweden's Prisma Satellites Accomplish Close Approach," *Spaceflight Now*, October 25, 2010, <https://spaceflightnow.com/news/n1010/25prisma>.
6. "Pair of Satellites Ejected from ISS for In-Space Navigation Exercise," *Spaceflight 101*, January 29, 2016, <https://spaceflight101.com/iss-expedition-46/pair-of-satellites-ejected-from-iss-for-in-space-navigation-exercise/>.
7. "OSAM-1: On-Orbit Servicing, Assembly, and Manufacturing-1," NASA's Exploration and In-space Services, accessed March 27, 2021, <https://nexus.gsfc.nasa.gov/OSAM-1.html>; Jeff Foust, "NASA reaffirms decision to cancel OSAM-1," *SpaceNews*, September 5, 2024, <https://spacenews.com/nasa-reaffirms-decision-to-cancel-osam-1/>.
8. "On-Orbit Servicing, Assembly, and Manufacturing-2 (OSAM-2)," NASA, last updated October 30, 2023, https://www.nasa.gov/mission_pages/tdm/osam-2.html.
9. "In-Orbit Now: Space Transportation. Reinvented," *D-Orbit SpA*, accessed March 28, 2021, <https://www.dorbit.space/>.
10. "Astroscale's ELSA-d Mission Successfully Completes Complex Rendezvous Operation," *Astroscale*, May 4, 2022, <https://astroscale.com/astrocales-elsa-d-mission-successfully-completes-complex-rendezvous-operation/>.
11. "A strong Public Interest and a Sustainable message for the future," *Cleospace Today*, accessed March 31, 2021, <https://cleospace.today/media/>.
12. "Clearspace-1," *The European Space Agency*, Accessed May 19, 2025, https://www.esa.int/Space_Safety/ClearSpace-1.
13. Debra Werner, "Major changes approved for ClearSpace-1 mission," *SpaceNews*, April 24, 2024, <https://spacenews.com/major-changes-approved-for-clearspace-1-mission/>.
14. Stephen Clark, "China says its launched a space debris mitigation tech demo satellite," *Spaceflight Now*, October 25, 2021, <https://spaceflightnow.com/2021/10/25/china-says-it-has-launched-a-space-debris-mitigation-tech-demo-satellite/>.
15. "Orbital Debris Quarterly News," *NASA Orbital Debris Program Office*, Volume 20, Issue 4, October 2016, <https://orbitaldebris.jsc.nasa.gov/quarterly-news/pdfs/odqnv20i4.pdf>.
16. Theresa Hitchens, "China's SJ-21 'tugs' dead satellite out of GEO belt: Trackers," *Breaking Defense*, January 26, 2022, <https://breakingdefense.com/2022/01/chinas-sj-21-tugs-dead-satellite-out-of-geo-belt-trackers/>.
17. "Astroscale's ADRAS-J Achieves Historic 15-Meter Approach to Space Debris," *Astroscale*, December 11, 2024, <https://astroscale.com/astrocales-adras-j-achieves-historic-15-meter-approach-to-space-debris/>.

18. "Northrop Grumman Successfully Completes Historic First Docking of Mission Extension Vehicle with Intelsat 901 Satellite," *Northrop Grumman*, February 26, 2020, <https://news.northropgrumman.com/news/releases/northrop-grumman-successfully-completes-historic-first-docking-of-mission-extension-vehicle-with-intelsat-901-satellite>; "Intelsat Completes Satellite Life-Extension Mission, Makes Space History," *Intelsat*, April 9, 2025, <https://www.intelsat.com/newsroom/intelsat-completes-satellite-life-extension-mission-makes-space-history/>; Andrew Jones, "2 private satellites undock after pioneering life-extension mission," *Space.com*, April 18, 2025, <https://www.space.com/space-exploration/satellites/2-private-satellites-undock-after-pioneering-life-extension-mission>.
19. "MEV-1,2," *Gunter's Space Page*, Accessed March 29, 2021, https://space.skyrocket.de/doc_sdat/mev-1.htm.
20. The Consortium for Execution of Rendezvous and Servicing Operations (CONFERS), <https://www.satelliteconfers.org/about-us>, accessed May 19, 2025.



525 Zang Street, STE. D
Broomfield, CO 80021 USA
v: + 1 303 554 1560

1779 Massachusetts Ave. NW
Washington, DC 20036 USA
v: + 1 202 568 6212

e: info@swfound.org